Validity Analysis: BLURB

Target context: EU/French Pharmaceutical Regulatory NLP — BLURB Assessment

Overall risk: **HIGH**

Deployment Context

Use case: In a pharmaceutical setting, a regulatory specialist uses document management software to ensure compliance in French drug labeling and scientific abstracts by applying a model evaluated on semantic textual similarity and fine-grained named-entity recognition. The system identifies over a dozen types of precise entities, such as chemical compounds and dosage modes, while verifying that safety warnings across different leaflets or patents are semantically consistent. These results determine whether pharmacological documentation meets strict professional standards for official submission or requires manual revision to prevent regulatory non-compliance.

Target population: Regulatory affairs specialists, pharmacologists, and legal experts at pharmaceutical companies or government health agencies.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology	2	Significant gaps	high
Input Content ⚠	1	Serious concern	high
Input Form ⚠	1	Serious concern	high
Output Ontology ⚠	2	Significant gaps	high
Output Content ⚠	2	Significant gaps	high
Output Form ✓	3	Moderate gaps	medium
Average	**1.8**		

⚠ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

BLURB is a well-constructed benchmark for English-language PubMed-based biomedical NLP, but it is fundamentally misaligned with the EU/French pharmaceutical regulatory NLP deployment across five of six validity dimensions. The language gap (English-only [Q1, Q126] vs French-language deployment), document-genre gap (PubMed abstracts vs SmPCs/PILs/CTD modules), entity-ontology gap (no INN/ATC/posology/excipient label spaces), STS-construct gap (general biomedical proximity [Q74] vs legally-determinative regulatory equivalence), and annotator-norm gap (biomedical researchers vs EMA/ANSM-aligned regulatory experts) are each rated HIGH severity by the elicitation and confirmed by web research [WEB-9, WEB-10, WEB-12, WEB-23]. Output form is the only dimension with reasonable representational alignment, and even there, the absence of calibration/threshold-sensitivity reporting limits direct utility for a human-review-triggering compliance pipeline. No remediation can be done within BLURB itself; bespoke French regulatory benchmark development against the EMA AI Reflection Paper [WEB-6] normative framework is required.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)
WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
WEB-23	Practitioner review confirms NLP for EU SmPC/prescribing information remains hybrid with no validated regulatory-specific benchmark (https://intuitionlabs.ai/articles/nlp-regulatory-labeling)

Practical Guidance

What This Benchmark Measures

BLURB measures the relative quality of English biomedical pretrained encoders on PubMed-derived NER, PICO extraction, relation extraction, sentence similarity (BIOSSES), document classification (HoC), and yes/no QA tasks [Q48, Q150]. For the EU/French regulatory deployment it can serve, at most, as an upper-bound sanity check on a French-translated or French-equivalent regulatory pipeline's underlying biomedical NLP competence — and even this only if a French biomedical PLM (DrBERT, CamemBERT-bio, AliBERT [WEB-14, WEB-15, WEB-16]) is the model under evaluation. None of the deployment's HIGH-priority constructs (regulatory NER ontology, regulatory equivalence STS, regulatory-expert annotation norms) is directly measurable through BLURB.

Construct Depth

BLURB probes generic biomedical NER, RE, STS, and QA constructs at moderate depth on English PubMed text. It does not probe regulatory-document understanding, French-language regulatory phrasing, regulatory-equivalence scoring, cross-document consistency, or calibration at decision boundaries. The five HIGH-priority dimensions in this deployment (IO, IC, IF, OO, OC) are essentially unaddressed by BLURB, and the MODERATE-priority OF dimension is only partially addressed (representation matches; calibration/threshold reporting does not).

What Else You Need

A complete assessment requires: (1) a French-language regulatory document corpus seeded by QUAERO EMEA [WEB-19, WEB-20] and informed by the DART Italian SmPC precedent [WEB-12]; (2) a regulatory-specific entity ontology (INN, ATC, excipient, posology, EMA-template qualifiers) with annotation guidelines anchored in EMA QRD v10.4 [WEB-1, WEB-2] and ANSM circulars; (3) a regulatory-equivalence STS dataset annotated by EMA/ANSM-trained regulatory affairs specialists, with critical-mismatch sub-categories (dose, population, indication scope); (4) calibration and threshold-sensitivity evaluation framed against the EMA AI Reflection Paper [WEB-6] and EU AI Act high-risk requirements; (5) an IAA study comparing biomedical NLP annotators with regulatory-expert annotators on shared regulatory text to quantify the OC gap.

ID	Evidence
Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-2	https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
WEB-14	DrBERT — French biomedical PLM pretrained from scratch (ACL 2023), not part of BLURB (https://arxiv.org/abs/2304.00958)
WEB-15	CamemBERT-bio — French biomedical continual-pretraining model (LREC-COLING 2024) (https://arxiv.org/abs/2306.15550)
WEB-16	AliBERT — French biomedical PLM with Unigram tokenizer (BioNLP 2023) (https://aclanthology.org/2023.bionlp-1.19/)
WEB-19	QUAERO French Medical Corpus — only public French regulatory-document NLP resource, very small EMEA subcorpus (https://quaerofrenchmed.limsi.fr/)
WEB-20	QUAERO HuggingFace dataset documentation — 4 EMEA documents segmented into 15 sub-documents (https://huggingface.co/datasets/DrBenchmark/QUAERO)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's task taxonomy (NER, PICO, relation extraction, sentence similarity, document classification, QA) is explicitly scoped to PubMed-based biomedical applications [Q46, Q48], with clinical and 'other high-value verticals' deferred to future work [Q46, Q197]. The deployment requires categories for regulatory document genres (SmPCs, PILs, CTD modules, patent claims) and regulatory-specific tasks such as cross-document consistency checking and regulatory equivalence STS — none of which are represented in BLURB's taxonomy. The QA scope is further restricted to yes/no and three-way classification [Q49, Q51, Q124], excluding the kind of structured field-extraction or compliance-verification task structure relevant to regulatory submissions. Web-confirmed: no public regulatory NLP benchmark with these task categories exists for French [WEB-12, WEB-23], reinforcing the structural gap. The taxonomy does include NER, document classification, and STS at an abstract level, which are the surface task types the deployment also uses, so the gap is one of category specificity rather than total absence.

ID	Evidence
Q46	"We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work." (p.6)
Q48	"BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering." (p.7)
Q49	"For question answering, prior work has considered both classification tasks (e.g., whether a reference text contains the answer to a given question) and more complex tasks such as list and summary [42]." (p.7)
Q51	"For simplicity, we focus on the classification tasks such as yes/no question-answering in BLURB, and leave the inclusion of more complex question-answering to future work." (p.7)
Q124	"For the two-way (yes/no) or three-way (yes/maybe/no) question-answering task, the encoding is similar to the sentence similarity task." (p.12)
Q197	"Future directions include: further exploration of domain-specific pretraining strategies; incorporating more tasks in biomedical NLP; extension of the BLURB benchmark to clinical and other high-value domains." (p.22)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
WEB-23	Practitioner review confirms NLP for EU SmPC/prescribing information remains hybrid with no validated regulatory-specific benchmark (https://intuitionlabs.ai/articles/nlp-regulatory-labeling)

Strengths

- BLURB's six-task structure (NER, RE, document classification, STS, QA, PICO) names the surface task types the deployment also relies on (NER, STS, document classification), providing a generic abstraction layer [Q48, Q195]
- The benchmark explicitly acknowledges that prior biomedical evaluations use heterogeneous task selections [Q37], and offers a stable task-grouped scoring scheme [Q52, Q150] that the deployment could in principle adopt for internal evaluation

ID	Evidence
Q37	"In contrast, prior work on biomedical pretraining tends to use different tasks and datasets for downstream evaluation, as shown in Table 2." (p.6)

Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q195	“To facilitate this study, we create BLURB, a comprehensive benchmark for biomedical NLP featuring a diverse set of tasks such as named entity recognition, relation extraction, document classification, and question answering.” (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for the deployment include regulatory NER (INNs, ATC codes, posology, excipients, EMA-template qualifiers), regulatory document classification (SmPC vs PIL vs CTD genre), regulatory equivalence STS (SmPC↔PIL, original↔revised), and cross-document consistency checking. None of these categories is present in BLURB; the closest analogs are generic biomedical NER, BIOSSES-style STS, and HoC-style abstract classification.

IO-2: Yes — BLURB omits regulatory document genres and regulatory-specific task types entirely. The benchmark explicitly defers 'other high-value verticals' to future work [Q46, Q197]. No public French or EU regulatory NLP benchmark covering these categories was identified [WEB-12, WEB-23].

IO-3: BLURB's PICO task (clinical trial P/I/O extraction) [Q59, Q111] and the molecular biology entity types in JNLPBA (DNA, RNA, cell line, cell type) [Q58] are not directly relevant to the regulatory submission deployment; they consume evaluation weight in the macro-averaged BLURB score [Q52, Q150] without contributing to regulatory-task signal.

IO-4: Content validity is harmed because the benchmark's task taxonomy underrepresents the regulatory-document task constructs the deployment needs to evaluate (genre-specific NER, regulatory equivalence, cross-document consistency) and includes research-oriented categories (PICO, molecular-biology NER) that do not generalize to SmPC/PIL/CTD prose.

ID	Evidence
Q46	“We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work.” (p.6)
Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q58	“JNLPBA. The Joint Workshop on Natural Language Processing in Biomedicine and its Applications shared task [27] is a NER corpus on PubMed abstracts. The entity types are chosen for molecular biology applications: protein, DNA, RNA, cell line, and cell type. Some of the entity type distinctions are not very meaningful. For example, a gene mention often refers to both the DNA and gene products such as the RNA and protein. Following prior work that evaluates on this dataset [34], we ignore the type distinction and focus on detecting the entity mentions. We use the same train, development, and test splits as in Crichton et al. [14].” (p.8)

Q59	“EBM PICO. The Evidence-Based Medicine corpus [44] contains PubMed abstracts on clinical trials, where each abstract is annotated with P, I, and O in PICO: Participants (e.g., diabetic patients), Intervention (e.g., insulin), Comparator (e.g., placebo) and Outcome (e.g., blood glucose levels). Comparator (C) labels are omitted as they are standard in clinical trials: placebo for passive control and standard of care for active control. There are 4300, 500, and 200 abstracts in training, development, and test, respectively.” (p.8)
Q111	“Conceptually, evidence-based medical information extraction is akin to slot filling, as it tries to identify the PIO elements in an abstract describing a clinical trial.” (p.11)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q197	“Future directions include: further exploration of domain-specific pretraining strategies; incorporating more tasks in biomedical NLP; extension of the BLURB benchmark to clinical and other high-value domains.” (p.22)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
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Information Gaps

- Whether any internal EMA tooling (Scientific Explorer, PICROSS, EurEKA) defines a public task taxonomy compatible with BLURB's framing

Requires Expert Verification

- The full set of regulatory task categories required for ANSM-compliant submission review (e.g., posology consistency, contraindication scope verification) — would benefit from regulatory affairs expert specification

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Remediation

Gap 1: BLURB's task taxonomy excludes regulatory document genres and regulatory-specific tasks (regulatory NER, regulatory equivalence STS, cross-document consistency).

Recommendation: Build a complementary task suite covering SmPC/PIL/CTD NER, regulatory equivalence STS (SmPC↔PIL, original↔revised), and EMA-template compliance classification. Use BLURB's six-task structure [Q48, Q52] as a reusable scoring scaffold but redefine the underlying task categories against EMA QRD v10.4 [WEB-1, WEB-2].

ID	Evidence
Q48	“BLURB is comprised of a comprehensive set of biomedical NLP tasks from publicly available datasets, including named entity recognition (NER), evidence-based medical information extraction (PICO), relation extraction, sentence similarity, document classification, and question answering.” (p.7)
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-2	https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's input content is drawn entirely from PubMed abstracts [Q1, Q53, Q54, Q56, Q57, Q58, Q59, Q64, Q70, Q126, Q127], with the pretraining corpus comprising 14 million abstracts [Q126]. The deployment targets French-language SmPCs, PILs, CTD modules, and patent claims — formulaic, template-governed regulatory documents that are categorically distinct from PubMed research prose. BLURB explicitly excludes even clinical notes on the grounds that they 'differ substantially from biomedical literature' [Q43]; this same logic applies a fortiori to regulatory documents, which differ from research abstracts in genre, register, legal function, and language. No French-language source, no SmPC, no PIL, and no EU regulatory submission document appears anywhere in BLURB's data. Web-confirmed: only the small QUAERO EMEA subcorpus (4 documents) provides any French regulatory NLP content [WEB-19, WEB-20], and even there the entity taxonomy is research-biomedical. The 2025 Frontiers SmPC IDMP study confirms empirically that even current LLMs miss ATC codes and excipient dosages in SmPC extraction [WEB-9], evidencing the magnitude of the content gap.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q43	“Clinical notes differ substantially from biomedical literature, to the extent that we observe BERT models pretrained on clinical notes perform poorly on biomedical tasks, similar to the standard BERT.” (p.6)
Q53	“The BioCreative V Chemical-Disease Relation corpus [35] was created for evaluating relation extraction of drug-disease interactions, but is frequently used as a NER corpus for detecting chemical (drug) and disease entities.” (p.7)
Q54	“The dataset consists of 1500 PubMed abstracts broken into three even splits for training, development, and test.” (p.7)
Q56	“NCBI-Disease. The Natural Center for Biotechnology Information Disease corpus [18] contains 793 PubMed abstracts with 6892 annotated disease mentions linked to 790 distinct disease entities. We use a pre-processed set of train, development, test splits generated by Crichton et al. [14].” (p.8)
Q57	“BC2GM. The Biocreative II Gene Mention corpus [53] consists of sentences from PubMed abstracts with manually labeled gene and alternative gene entities. Following prior work, we focus on the gene entity annotation. In its original form, BC2GM contains 15000 train and 5000 test sentences. We use a pre-processed version of the dataset generated by Crichton et al. [14], which carves out 2500 sentences from the training data for development.” (p.8)
Q58	“JNLPBA. The Joint Workshop on Natural Language Processing in Biomedicine and its Applications shared task [27] is a NER corpus on PubMed abstracts. The entity types are chosen for molecular biology applications: protein, DNA, RNA, cell line, and cell type. Some of the entity type distinctions are not very meaningful. For example, a gene mention often refers to both the DNA and gene products such as the RNA and protein. Following prior work that evaluates on this dataset [34], we ignore the type distinction and focus on detecting the entity mentions. We use the same train, development, and test splits as in Crichton et al. [14].” (p.8)
Q59	“EBM PICO. The Evidence-Based Medicine corpus [44] contains PubMed abstracts on clinical trials, where each abstract is annotated with P, I, and O in PICO: Participants (e.g., diabetic patients), Intervention (e.g., insulin), Comparator (e.g., placebo) and Outcome (e.g., blood glucose levels). Comparator (C) labels are omitted as they are standard in clinical trials: placebo for passive control and standard of care for active control. There are 4300, 500, and 200 abstracts in training, development, and test, respectively.” (p.8)
Q64	“ChemProt. The Chemical Protein Interaction corpus [31] consists of PubMed abstracts annotated with chemical-protein interactions between chemical and protein entities. There are 23 interactions organized in a hierarchy, with 10 high-level interactions (including NONE). The vast majority of relation instances in ChemProt are within single sentences. Following prior work [8, 34], we only consider sentence-level instances.” (p.8)
Q70	“The Genetic Association Database corpus [11] was created semi-automatically using the Genetic Association Archive.” (p.9)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)

Q127	“The original collection contains over 4 billion words; we filter out any abstracts with less than 128 words to reduce noise.” (p.12)
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)
WEB-19	QUAERO French Medical Corpus — only public French regulatory-document NLP resource, very small EMEA subcorpus (https://quaerofrenchmed.limsi.fr/)
WEB-20	QUAERO HuggingFace dataset documentation — 4 EMEA documents segmented into 15 sub-documents (https://huggingface.co/datasets/DrBenchmark/QUAERO)

Strengths

- Chemical-entity content in BC5-Chemical and ChemProt [Q53, Q64] partially overlaps with active-substance vocabulary that also appears in regulatory text, providing some lexical transfer signal for INN-adjacent entities
- The DDI corpus [Q68] addresses pharmacovigilance-adjacent content ('drug-drug interactions ... particular focus on pharmacovigilance'), which is the closest BLURB content to a regulatory information-extraction concern

ID	Evidence
Q53	“The BioCreative V Chemical-Disease Relation corpus [35] was created for evaluating relation extraction of drug-disease interactions, but is frequently used as a NER corpus for detecting chemical (drug) and disease entities.” (p.7)
Q64	“ChemProt. The Chemical Protein Interaction corpus [31] consists of PubMed abstracts annotated with chemical-protein interactions between chemical and protein entities. There are 23 interactions organized in a hierarchy, with 10 high-level interactions (including NONE). The vast majority of relation instances in ChemProt are within single sentences. Following prior work [8, 34], we only consider sentence-level instances.” (p.8)
Q68	“DDI. The Drug-Drug Interaction corpus [21] was created to facilitate research on pharmaceutical information extraction, with a particular focus on pharmacovigilance. It contains sentence-level annotation of drug-drug interactions on PubMed abstracts. Note that some prior work [45, 61] discarded 90 training files that the authors” (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — the deployment requires French-language regulatory phrasing, EMA QRD template language [WEB-1, WEB-2], ANSM-jurisdiction conventions, French posology expressions ('deux comprimés par jour'), and EU-specific nomenclature (E-numbers, Ph. Eur. terms). None of this is present in BLURB's English PubMed corpus [Q1, Q126].

IC-2: Not aligned. BLURB content is research-biomedical English prose [Q1]; deployment culture is EU/French regulatory-legal. The content register (formulaic, template-driven) and language are both mismatched.

IC-3: Yes — BLURB inputs are entirely English PubMed abstracts [Q126, Q127] reflecting Anglophone biomedical research conventions. The benchmark itself notes that 'general domain text is substantively different from biomedical text' [Q12]; the same mismatch exists between PubMed abstracts and EU regulatory documents.

IC-4: INSUFFICIENT DOCUMENTATION — would require recruitment of French regulatory affairs specialists to review BLURB samples for relevance to SmPC/PIL/CTD content; not addressed in the paper.

IC-5: Content validity is severely harmed: every BLURB datapoint is in the wrong language, wrong document genre, and wrong register relative to the deployment. Construct-irrelevant variance is essentially total at the content level.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q12	“In fact, the majority of general domain text is substantively different from biomedical text, raising the prospect of negative transfer that actually hinders the target performance.” (p.2)
Q43	“Clinical notes differ substantially from biomedical literature, to the extent that we observe BERT models pretrained on clinical notes perform poorly on biomedical tasks, similar to the standard BERT.” (p.6)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
Q127	“The original collection contains over 4 billion words; we filter out any abstracts with less than 128 words to reduce noise.” (p.12)
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)
WEB-2	https://www.ema.europa.eu/en/documents/template-form/qrd-product-information-template-version-104-highlighted_en.pdf
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)

Information Gaps

- Quantification of lexical overlap between BLURB chemical/drug vocabulary and INN/ATC nomenclature for the deployment's drug list

Requires Expert Verification

- Whether the QUAERO EMEA subcorpus is fit-for-purpose as a domain-adaptation seed for the deployment

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Remediation

Gap 1: BLURB contains zero French-language and zero regulatory-document content; all inputs are English PubMed abstracts [Q1, Q126].

Recommendation: Construct a French regulatory corpus using the QUAERO EMEA subcorpus [WEB-19, WEB-20] as a seed, supplemented by EMA-published SmPCs/PILs and ANSM-published French regulatory texts. Treat the DART Italian SmPC corpus [WEB-12] as a development-roadmap precedent.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)
WEB-19	QUAERO French Medical Corpus — only public French regulatory-document NLP resource, very small EMEA subcorpus (https://quaerofrenchmed.limsi.fr/)
WEB-20	QUAERO HuggingFace dataset documentation — 4 EMEA documents segmented into 15 sub-documents (https://huggingface.co/datasets/DrBenchmark/QUAERO)

Input Form (IF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB is exclusively English-language text [Q1, Q46, Q126], with vocabulary generated from PubMed [Q126] and tokenization optimized for English biomedical terminology [Q29, Q162, Q168]. The deployment operates on French-language regulatory documents with French diacritics and French-specific tokenization needs. No multilingual processing, no French-language variant, and no porting strategy is documented in BLURB. The benchmark's own argument that vocabulary mismatch causes performance degradation — 'standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords' [Q28], with examples like 'naloxone' fragmenting into four pieces [Q29] — applies with even greater force across the English↔French boundary. Sequence-length truncations (128/256/512 tokens [Q120, Q125]) were calibrated for PubMed sentence/abstract structure, not for SmPC sections or PIL prose. Web-confirmed French biomedical models (DrBERT, CamemBERT-bio, AliBERT) [WEB-14, WEB-15, WEB-16, WEB-17] exist but are not part of BLURB and have not been fine-tuned on EMA/ANSM regulatory genres. Form mismatch is foundational.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q28	“As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords.” (p.5)
Q29	“For example, naloxone, a common medical term, is divided into four pieces ([na, ##lo, ##xon, ##e]) by BERT, and acetyltransferase is shattered into seven pieces ([ace, ##ty, ##lt, ##ran, ##sf, ##eras, ##e]) by BERT.” (p.5)
Q46	“We focus on PubMed-based biomedical applications, and leave the exploration of the clinical domain, and other high-value verticals to future work.” (p.6)
Q120	“For computational efficiency, we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI which contain longer input sequences.” (p.12)
Q125	“For computational efficiency, we use padding or truncation to set the input length to 512 tokens.” (p.12)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
Q162	“A significant advantage in using an in-domain vocabulary is that the input will be shorter in downstream tasks, as shown in Table 8, which makes learning easier.” (p.15)
Q168	“Table 8. Comparison of the average input length in word pieces using general-domain vs in-domain vocabulary.” (p.17)
WEB-14	DrBERT — French biomedical PLM pretrained from scratch (ACL 2023), not part of BLURB (https://arxiv.org/abs/2304.00958)
WEB-15	CamemBERT-bio — French biomedical continual-pretraining model (LREC-COLING 2024) (https://arxiv.org/abs/2306.15550)
WEB-16	AliBERT — French biomedical PLM with Unigram tokenizer (BioNLP 2023) (https://aclanthology.org/2023.bionlp-1.19/)
WEB-17	DrBenchmark — 20-task French biomedical evaluation suite, separate from BLURB and not regulatory-specific (https://aclanthology.org/2024.lrec-main.478/)

Strengths

- BLURB establishes that in-domain vocabulary and tokenization materially affect performance [Q25, Q28, Q160, Q162], providing a transferable methodological lesson the deployment can apply when training a French regulatory tokenizer

• Standard transformer preprocessing scaffolding (special token insertion, padding/truncation, entity dummification) [Q99, Q120, Q183, Q184] is technology-neutral with respect to language and could be reused in a French regulatory NLP pipeline

ID	Evidence
Q25	“A major advantage of domain-specific pretraining from scratch stems from having an in-domain vocabulary.” (p.5)
Q28	“As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords.” (p.5)
Q99	“An input instance is first processed by a TransformInput module which performs task-specific transformations such as appending special instance marker (e.g., [CLS]) or dummifying entity mentions for relation extraction.” (p.10)
Q120	“For computational efficiency, we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI which contain longer input sequences.” (p.12)
Q160	“Using the original BERT vocabulary derived from Wikipedia & BookCorpus (by continual pretraining from the original BERT), the results are significantly worse than using an in-domain vocabulary from PubMed.” (p.15)
Q162	“A significant advantage in using an in-domain vocabulary is that the input will be shorter in downstream tasks, as shown in Table 8, which makes learning easier.” (p.15)
Q183	“With entity dummification, the entity mentions in question are anonymized using entity type tags such as $DRUG or $GENE.” (p.20)
Q184	“With entity marker, special tags marking the start and end of an entity are appended to the entity mentions in question.” (p.20)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions are mismatched: BLURB inputs are English PubMed text tokenized with a PubMed-derived vocabulary [Q126], whereas the deployment requires French regulatory text with French diacritics (é, è, ê, ç, œ) and regulatory-specific tokens (INNs, ATC codes, posology phrases). The benchmark's own demonstration that vocabulary fragmentation harms performance [Q28, Q29] applies to this mismatch.

IF-2: Infrastructure is not the constraint — the deployment population has full enterprise infrastructure. The form mismatch is at the linguistic/encoding level, not the capture level.

IF-3: Yes: regulatory documents have template-driven structure (numbered SmPC sections, PIL headings) that BLURB's flat-abstract preprocessing [Q84] does not model. Sequence-length caps of 128/256/512 tokens [Q120, Q125] may also truncate long SmPC sections or cross-section equivalence pairs.

IF-4: External validity is severely harmed: the benchmark cannot evaluate model behavior on French text, French regulatory tokenization, or template-structured documents — precisely the input form the deployment must handle.

ID	Evidence
Q1	“we compile a comprehensive biomedical NLP benchmark from publicly-available datasets.” (p.1)
Q28	“As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords.” (p.5)
Q29	“For example, naloxone, a common medical term, is divided into four pieces ([na, ##lo, ##xon, ##e]) by BERT, and acetyltransferase is shattered into seven pieces ([ace, ##ty, ##lt, ##ran, ##sf, ##eras, ##e]) by BERT.” (p.5)
Q84	“Though the original dataset provided sentence level annotation, we follow the common practice and evaluate on the abstract level [19, 60].” (p.9)

Q120	“For computational efficiency, we use padding or truncation to set the input length to 128 tokens for GAD and 256 tokens for ChemProt and DDI which contain longer input sequences.” (p.12)
Q125	“For computational efficiency, we use padding or truncation to set the input length to 512 tokens.” (p.12)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
WEB-14	DrBERT — French biomedical PLM pretrained from scratch (ACL 2023), not part of BLURB (https://arxiv.org/abs/2304.00958)
WEB-17	DrBenchmark — 20-task French biomedical evaluation suite, separate from BLURB and not regulatory-specific (https://aclanthology.org/2024.lrec-main.478/)

Information Gaps

- Whether SmPC/PIL section length distributions exceed BLURB's 512-token cap in practice

Requires Expert Verification

- Optimal French regulatory tokenizer construction strategy (vocabulary scope, treatment of INNs/ATC codes as multi-word units)

Remediation

Gap 1: BLURB is exclusively English with PubMed-derived tokenization [Q126]; the deployment requires French tokenization with regulatory vocabulary support.

Recommendation: Adopt a French biomedical PLM (DrBERT, CamemBERT-bio, or AliBERT [WEB-14, WEB-15, WEB-16]) as the encoder and extend its tokenizer with regulatory-specific tokens (INNs, ATC codes, French posology multi-word units), applying BLURB's own in-domain-vocabulary lessons [Q25, Q28].

ID	Evidence
Q25	“A major advantage of domain-specific pretraining from scratch stems from having an in-domain vocabulary.” (p.5)
Q28	“As a result, standard BERT models are forced to divert parametrization capacity and training bandwidth to model biomedical terms using fragmented subwords.” (p.5)
Q126	“For biomedical domain-specific pretraining, we generate the vocabulary and conduct pretraining using the latest collection of PubMed abstracts: 14 million abstracts, 3.2 billion words, 21 GB.” (p.12)
WEB-14	DrBERT — French biomedical PLM pretrained from scratch (ACL 2023), not part of BLURB (https://arxiv.org/abs/2304.00958)
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WEB-16	AliBERT — French biomedical PLM with Unigram tokenizer (BioNLP 2023) (https://aclanthology.org/2023.bionlp-1.19/)

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's output ontology consists of task-specific label spaces: BIO/BIOUL token tags for NER over chemical/disease/gene/molecular-biology entity types [Q107, Q108, Q109], five-way ChemProt relation classes [Q65], yes/no and yes/maybe/no QA labels [Q86, Q90], binary cancer hallmark indicators [Q78, Q80], and a 0–4 BIOSSES regression score [Q74, Q75]. None of these label spaces accommodates regulatory-specific entity types (INNs, ATC codes, excipient nomenclature, posology expressions, EMA-template contraindication qualifiers) or regulatory-equivalence scoring. The BIOSSES regression operationalizes general biomedical semantic proximity ['estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)' Q74] — not the legally determinative dose-threshold/population-qualifier sensitivity the deployment requires. Web-confirmed: no regulatory-equivalence STS benchmark exists [WEB-10], and 2025 SmPC LLM extraction shows ATC codes and excipient dosages are frequently misclassified [WEB-9]. The macro-averaged BLURB summary score [Q52, Q150] does not encode any decision-boundary or human-review-trigger semantics. Some weak structural overlap exists (binary classification, regression scoring) at the abstract level, hence score 2 rather than 1.

ID	Evidence
Q52	"To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types." (p.7)
Q65	"We follow the ChemProt authors' suggestions and focus on classifying five high-level interactions — UPREGULATOR (CPR : 3), DOWNREGULATOR (CPR : 4), AGONIST (CPR : 5), ANTAGONIST (CPR : 6), SUBSTRATE (CPR : 9) — as well as everything else (false)." (p.8)
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
Q75	"It is a regression task, with the average score as the final annotation." (p.9)
Q78	"It contains annotation on PubMed abstracts with binary labels each of which signifies the discussion of a specific cancer hallmark." (p.9)
Q80	"We focus on predicting the top-level labels." (p.9)
Q86	"The PubMedQA dataset [25] contains a set of research questions, each with a reference text from a PubMed abstract as well as an annotated label of whether the text contains the answer to the research question (yes/maybe/no)." (p.9)
Q90	"Each question is paired with a reference text containing multiple sentences from a PubMed abstract and a yes/no answer." (p.9)
Q107	"BIO is the standard tagging scheme that classifies each token as the beginning of an entity (B), inside an entity (I), or outside (O)." (p.11)
Q108	"The NER tasks in BLURB are only concerned about one entity type (in JNLPBA, all the types are merged into one)." (p.11)
Q109	"Prior work has also considered more complex tagging schemes such as BIOUL, where U stands for the last word of an entity and L stands for a single-word entity." (p.11)
Q150	"The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering)." (p.14)
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)

WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)

Strengths

- BLURB demonstrates that task-specific prediction layers can be cleanly layered atop a shared encoder [Q93, Q94, Q103, Q104], a transferable architectural pattern for regulatory tasks
- BIOSSES provides a concrete sentence-similarity regression schema with expert-derived gold scores [Q74, Q75] that could be structurally adapted into a regulatory-equivalence schema (with re-defined anchors)
- ChemProt's hierarchical relation taxonomy [Q65] is closer to a structured pharmaceutical interaction schema than other BLURB tasks

ID	Evidence
Q65	"We follow the ChemProt authors' suggestions and focus on classifying five high-level interactions — UPREGULATOR (CPR : 3), DOWNREGULATOR (CPR : 4), AGONIST (CPR : 5), ANTAGONIST (CPR : 6), SUBSTRATE (CPR : 9) — as well as everything else (false)." (p.8)
Q74	"The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings)." (p.9)
Q75	"It is a regression task, with the average score as the final annotation." (p.9)
Q93	"Given an input token sequence, the language model produces a sequence of vectors in the contextual representation." (p.9)
Q94	"A task-specific prediction model is then layered on top to generate the final output for a task-specific application." (p.9)
Q103	"Many NLP applications can be formulated as a classification or regression task, wherein either individual tokens or sequences are the prediction target." (p.11)
Q104	"Modeling choices usually vary in two aspects: the instance representation and the prediction layer." (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output label categories are research-biomedical: chemical/disease/gene NER tags [Q108], generic chemical-protein relations [Q65], yes/no QA [Q86, Q90], cancer hallmarks [Q78]. None directly map onto regulatory NER categories (INNs, ATC, excipients, posology) or regulatory equivalence categories (dose-threshold mismatch, population-qualifier mismatch).

OO-2: Missing categories: INN entity type, ATC code entity type, excipient entity type, posology expression entity type, EMA-template contraindication qualifier; and on the STS side, regulatory equivalence with critical-mismatch sub-categories (dose, population, indication scope). The web search confirms no public benchmark covers these for French [WEB-10, WEB-12].

OO-3: BLURB's BIOSSES regression encodes 'general semantic proximity' [Q74] as the construct, which is value-laden in the sense that it weights paraphrase/synonymy uniformly — incompatible with the deployment's principle that small lexical differences (e.g., dose thresholds) carry legal weight disproportionate to their lexical distance.

OO-4: A stakeholder-driven taxonomy redesign is required: regulatory affairs experts under EMA/ANSM/EU AI Act normative framing [WEB-6, WEB-11] would need to define both the entity ontology and the equivalence-class taxonomy.

OO-5: Structural validity is harmed (the construct's structure — regulatory equivalence as legally-weighted proximity — is misrepresented as uniform semantic proximity); content validity is harmed (regulatory entity categories absent); external validity is at risk because BLURB scores cannot be expected to track regulatory compliance performance.

ID	Evidence
Q65	“We follow the ChemProt authors' suggestions and focus on classifying five high-level interactions — UPREGULATOR (CPR : 3), DOWNREGULATOR (CPR : 4), AGONIST (CPR : 5), ANTAGONIST (CPR : 6), SUBSTRATE (CPR : 9) — as well as everything else (false).” (p.8)
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)
Q75	“It is a regression task, with the average score as the final annotation.” (p.9)
Q78	“It contains annotation on PubMed abstracts with binary labels each of which signifies the discussion of a specific cancer hallmark.” (p.9)
Q86	“The PubMedQA dataset [25] contains a set of research questions, each with a reference text from a PubMed abstract as well as an annotated label of whether the text contains the answer to the research question (yes/maybe/no).” (p.9)
Q90	“Each question is paired with a reference text containing multiple sentences from a PubMed abstract and a yes/no answer.” (p.9)
Q108	“The NER tasks in BLURB are only concerned about one entity type (in JNLPBA, all the types are merged into one).” (p.11)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)
WEB-9	2025 Frontiers SmPC IDMP study — ATC codes and excipient dosages 'frequently missing or misclassified' by LLMs (https://www.frontiersin.org/journals/medicine/articles/10.3389/fmed.2025.1598979/full)
WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)
WEB-11	EMA AI page including LLM Guiding Principles (Sep 2024) (https://www.ema.europa.eu/en/about-us/how-we-work/data-regulation-big-data-other-sources/artificial-intelligence)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)

Information Gaps

- Whether the deployment's regulatory equivalence scoring should be ordinal, multi-label (per critical mismatch category), or hierarchical

Requires Expert Verification

- Definition of EMA/ANSM-aligned critical-mismatch categories and their decision thresholds for human-review triggering

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Remediation

Gap 1: No label space for regulatory entity types (INN, ATC, excipient, posology, EMA-template qualifiers); BIOSSES regression encodes general semantic proximity, not regulatory equivalence [Q74, Q108].

Recommendation: Define a regulatory entity ontology aligned with WHO INN/ATC and EDQM Standard Terms, and define a regulatory-equivalence STS schema with explicit critical-mismatch sub-categories (dose threshold, contraindicated population, indication scope), informed by the deployment's documented critical mismatch list and the EMA AI Reflection Paper [WEB-6].

ID	Evidence
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)
Q108	“The NER tasks in BLURB are only concerned about one entity type (in JNLPBA, all the types are merged into one).” (p.11)

WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's annotator population is biomedical-research-oriented and heterogeneous: Amazon Mechanical Turkers and Upwork contributors with medical training for EBM PICO [Q60], 'five expert-level annotators' for BIOSSES [Q74], 'biomedical experts' for BioASQ [Q88], and unspecified/varied for the remaining datasets. None are regulatory affairs specialists or legal experts operating under EMA/ANSM normative frameworks. The deployment explicitly anticipates that biomedical NLP annotators will 'prioritize clinical relevance' while regulatory ground truth requires 'rigid legal constraint satisfaction,' producing systematic disagreements on borderline cases (elicitation A4). Web-confirmed: no published IAA study comparing biomedical vs regulatory annotators exists, and no regulatory-expert-annotated French pharmaceutical NER/STS dataset has been identified [WEB-12]. BLURB also does not systematically report IAA, annotator demographics, or annotation protocols across datasets. ChemProt label expansion via assigning false labels to unannotated pairs [Q66] further reflects research-pragmatic conventions that may be inappropriate where unannotated content has legal-evidentiary weight. Some annotation quality signals (BIOSSES expert annotators, EBM PICO test set quality stratification) [Q60, Q74] exist, hence score 2 rather than 1.

ID	Evidence
Q60	“The training and development sets were labeled by Amazon Mechanical Turkers, whereas the test set was labeled by Upwork contributors with prior medical training.” (p.8)
Q66	“The ChemProt annotation is not exhaustive for all chemical-protein pairs. Following previous work [34, 45], we expand the training and development sets by assigning a false label for all chemical-protein pairs that occur in a training or development sentence, but do not have an explicit label in the ChemProt corpus.” (p.8)
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)
Q88	“The BioASQ corpus [42] contains multiple question answering tasks annotated by biomedical experts, including yes/no, factoid, list, and summary questions.” (p.9)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)

Strengths

- BIOSSES uses five expert annotators with documented inter-annotator process [Q74], offering a usable methodological template for setting up regulatory-expert annotation
- BLURB documents the EBM PICO quality stratification (MTurk train/dev vs medically trained Upwork test) [Q60], demonstrating awareness that annotator expertise affects label quality
- BioASQ annotations by 'biomedical experts' [Q88] provide a partial precedent for domain-expert annotation in QA construction

ID	Evidence
Q60	“The training and development sets were labeled by Amazon Mechanical Turkers, whereas the test set was labeled by Upwork contributors with prior medical training.” (p.8)
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)

Q88	“The BioASQ corpus [42] contains multiple question answering tasks annotated by biomedical experts, including yes/no, factoid, list, and summary questions.” (p.9)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground truth labels do not reflect regional regulatory stakeholder perspectives. Annotators are biomedical/clinical professionals [Q60, Q74, Q88], not regulatory affairs specialists or EMA/ANSM-aligned legal experts.

OC-2: Disagreement is explicitly anticipated by the deployment elicitation (A4) on borderline cases, where biomedical annotators prioritize clinical relevance while regulatory ground truth prioritizes legal constraint satisfaction. No empirical IAA study addresses this gap [WEB-12].

OC-3: Annotator demographics are unevenly documented: EBM PICO sources are named [Q60], BIOSSES annotator count and expertise level given [Q74], BioASQ described generically as 'biomedical experts' [Q88]; for most datasets (BC5, NCBI-Disease, BC2GM, JNLPBA, ChemProt, DDI, GAD, HoC, PubMedQA) the paper does not provide annotator demographics or protocols. INSUFFICIENT DOCUMENTATION across most of the benchmark.

OC-4: Re-annotation by EMA/ANSM-trained regulatory affairs specialists would be required for any deployment use, but is not feasible without a French regulatory document corpus to begin with.

OC-5: BLURB uses macro-averaging across task types [Q52, Q150] and averages across multiple runs for small datasets [Q139, Q140] but does not document any aggregation method that preserves minority annotator perspectives — particularly relevant for the BIOSSES five-annotator average [Q75], which collapses disagreement into a single regression target.

OC-6: Convergent validity is harmed (labels do not correlate with regulatory-stakeholder judgments); external validity is harmed (BLURB performance is unlikely to predict performance against EMA/ANSM ground truth).

ID	Evidence
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q60	“The training and development sets were labeled by Amazon Mechanical Turkers, whereas the test set was labeled by Upwork contributors with prior medical training.” (p.8)
Q74	“The Sentence Similarity Estimation System for the Biomedical Domain [54] contains 100 pairs of PubMed sentences each of which is annotated by five expert-level annotators with an estimated similarity score in the range from 0 (no relation) to 4 (equivalent meanings).” (p.9)
Q75	“It is a regression task, with the average score as the final annotation.” (p.9)
Q88	“The BioASQ corpus [42] contains multiple question answering tasks annotated by biomedical experts, including yes/no, factoid, list, and summary questions.” (p.9)
Q139	“Due to random initialization of the task-specific model and drop out, the performance may vary for different random seeds, especially for small datasets like BIOSSES, BioASQ, and PubMedQA.” (p.13)
Q140	“We report the average scores from ten runs for BIOSSES, BioASQ, and PubMedQA, and five runs for the others.” (p.13)

Q150	"The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering)." (p.14)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)
WEB-10	EMA 2024 AI Observatory Report — internal regulatory NLP tools developed under EMA staff oversight, not represented in any public benchmark (https://www.ema.europa.eu/en/documents/report/2024-ai-observatory-report_en.pdf)
WEB-12	No regulatory-expert-annotated pharmaceutical NLP dataset (French or English) identified; DART Italian corpus does not use regulatory-expert annotators (https://arxiv.org/html/2510.18475)

Information Gaps

- Empirical magnitude of disagreement between biomedical and regulatory annotators on shared regulatory text — no IAA study exists [WEB-12]

Requires Expert Verification

- Whether existing EMA/ANSM internal review checklists could be operationalized as annotation guidelines for a regulatory-grade gold standard

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Remediation

Gap 1: No BLURB annotators are regulatory affairs specialists or EMA/ANSM-aligned legal experts; systematic disagreement on borderline cases is anticipated (elicitation A4).

Recommendation: Establish a regulatory-expert annotation pool drawn from regulatory affairs and pharmacovigilance specialists, with annotation guidelines anchored in EMA QRD templates [WEB-1] and ANSM procedural documentation. Run a calibrated IAA study comparing regulatory-expert vs biomedical-NLP annotator labels on a shared sample of regulatory text to quantify the gap.

ID	Evidence
WEB-1	EMA QRD SmPC Template v10.4 (Feb 2024) — defines the formulaic template language the deployment must process (https://www.ema.europa.eu/en/human-regulatory-overview/marketing-authorisation/product-information-requirements/product-information-qrd-templates-human)

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: *BLURB* | ***Context:*** *EU/French Pharmaceutical Regulatory NLP — BLURB Assessment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

BLURB's output forms — categorical labels for classification, BIO/BIOUL sequence tags for NER, regression scores for STS, accuracy for QA, F1 variants for classification/extraction, Pearson correlation for STS [Q76, Q83, Q100, Q176] — broadly match the representational forms the deployment requires (entity tags, similarity/equivalence scores, classification labels). At this representational level the form mismatch is modest, and the deployment elicitation rates OF as MODERATE priority. However, BLURB reports only aggregate metrics: macro-averaged BLURB score [Q52, Q150], task-level F1/accuracy/Pearson, and averages across multiple runs for variance management [Q140]. There are no confidence-calibration metrics, threshold-sensitivity analyses, or human-review-triggering decision boundaries documented [Q150, Q151, Q152] — precisely the output-form information the deployment's human-review flagging workflow requires. The deployment's binary 'flag for human review' decision derives from continuous model outputs, but BLURB does not characterize model behavior at decision boundaries or report calibration. The form match is therefore partial: representational forms align, but evaluation forms (calibration, threshold sensitivity) do not.

ID	Evidence
Q52	“To compute a summary score for BLURB, the simplest way is to report the average score among all tasks. However, this may place undue emphasis on simpler tasks such as NER for which there are many existing datasets. Therefore, we group the datasets by their task types, compute the average score for each task type, and report the macro average among the task types.” (p.7)
Q76	“We use the same train/dev/test split in Peng et al. [45] and use Pearson correlation for evaluation.” (p.9)
Q83	“We instead adopt the original dataset and report micro F1 across the ten cancer hallmarks.” (p.9)
Q100	“We use cross-entropy loss for classification tasks and mean-square error for regression tasks.” (p.10)
Q140	“We report the average scores from ten runs for BIOSSES, BioASQ, and PubMedQA, and five runs for the others.” (p.13)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q151	“We compare BERT models by applying them to the downstream NLP applications in BLURB.” (p.14)
Q152	“For each task, we conduct the same fine-tuning process using the standard task-specific model as specified in subsection 2.4.” (p.14)
Q176	“Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes and the standard PubMedBERT.” (p.19)

Strengths

- Output representational forms (categorical labels, sequence tags, regression scores) [Q100, Q107, Q176] directly map to the deployment's NER and STS output requirements
- BLURB provides standardized task-specific metrics (entity-level F1, micro F1, Pearson) [Q76, Q83, Q176] that are widely used and interpretable, easing methodological reuse
- Public release of pretrained and task-specific models alongside the leaderboard [Q3, Q196] supports reproducibility, which is valuable for any regulatory NLP system that must be auditable

ID	Evidence

Q3	“we have released our state-of-the-art pretrained and task-specific models for the community, and created a leaderboard featuring our BLURB benchmark (short for Biomedical Language Understanding & Reasoning Benchmark).” (p.1)
Q76	“We use the same train/dev/test split in Peng et al. [45] and use Pearson correlation for evaluation.” (p.9)
Q83	“We instead adopt the original dataset and report micro F1 across the ten cancer hallmarks.” (p.9)
Q100	“We use cross-entropy loss for classification tasks and mean-square error for regression tasks.” (p.10)
Q107	“BIO is the standard tagging scheme that classifies each token as the beginning of an entity (B), inside an entity (I), or outside (O).” (p.11)
Q176	“Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes and the standard PubMedBERT.” (p.19)
Q196	“To accelerate research in biomedical NLP, we release our state-of-the-art biomedical BERT models and setup a leaderboard based on BLURB.” (p.21)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text labels, scores) matches the deployment's enterprise document-management integration; no speech, no mobile-output requirement is in scope. Representational match is good.

OF-2: Not applicable — the deployment is a desktop/server enterprise context with no speech-output requirement.

OF-3: Not applicable — end users are university-educated regulatory professionals; literacy and accessibility are not constraints.

OF-4: External validity is harmed at the evaluation-form level: BLURB does not report confidence calibration or threshold-sensitivity analyses [Q150, Q151], and the human-review flagging workflow requires precisely this information. Aggregate F1/Pearson scores do not characterize model behavior at the decision boundary the deployment uses.

ID	Evidence
Q100	“We use cross-entropy loss for classification tasks and mean-square error for regression tasks.” (p.10)
Q107	“BIO is the standard tagging scheme that classifies each token as the beginning of an entity (B), inside an entity (I), or outside (O).” (p.11)
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q151	“We compare BERT models by applying them to the downstream NLP applications in BLURB.” (p.14)
Q176	“Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes and the standard PubMedBERT.” (p.19)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)

Information Gaps

- Whether the deployment's human-review flagging is implemented as a fixed score threshold or a calibrated probability cutoff

Requires Expert Verification

- Calibration and threshold-sensitivity reporting requirements implied by EU AI Act high-risk classification for the deployment

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Remediation

Gap 1: BLURB reports only aggregate metrics (macro-averaged scores, F1, Pearson) [Q150, Q176] and no calibration or threshold-sensitivity analysis, but the deployment requires confidence-based human-review triggering.

Recommendation: Augment evaluation with calibration metrics (ECE, reliability diagrams), threshold-sensitivity sweeps, and decision-boundary characterization against a regulatory-expert gold standard. Document these in line with EMA AI Reflection Paper expectations [WEB-6] and EU AI Act high-risk-system requirements.

ID	Evidence
Q150	“The BLURB score is the macro average of average test results for each of the six tasks (NER, PICO, relation extraction, sentence similarity, document classification, question answering).” (p.14)
Q176	“Comparison of entity-level F1 for biomedical named entity recognition (NER) using different tagging schemes and the standard PubMedBERT.” (p.19)
WEB-6	EMA AI Reflection Paper requires human-oversight mechanisms and quality review for AI in product information drafting — implies threshold/calibration characterization the deployment must satisfy (https://www.ema.europa.eu/en/documents/scientific-guideline/reflection-paper-use-artificial-intelligence-ai-medicinal-product-lifecycle_en.pdf)